

Classification of Surgical Specialty Categories

Overview: We categorized surgical procedures according to major clinical specialties commonly used in surgical research. Each specialty grouping reflects the dominant anatomic region, operative approach, or discipline-specific expertise described in the source studies.

Category 1 — General Surgery

Definition: Encompasses abdominal, gastrointestinal, hepatobiliary, pancreatic, colorectal, esophageal, gastric, bowel, splenic, and hepatic procedures, as well as transplant surgery and vascular operations such as EVAR, TEVAR, abdominal aortic aneurysm repair, and aortoiliac reconstruction.

Category 2 — Thoracic Surgery

Definition: Covers non-cardiac thoracic procedures, including lung resection, lobectomy, segmentectomy, pulmonary resection, thoracotomy, and video-assisted thoracoscopic surgery (VATS).

Category 3 — Cardiac Surgery

Definition: Includes coronary artery bypass grafting (CABG), valve surgery, aortic surgery, open-heart procedures, and other forms of coronary revascularization.

Category 4 — Orthopedic / Spine Surgery

Definition: Musculoskeletal and spine procedures such as hip fracture repair, joint arthroplasty, and spinal operations including anterior cervical discectomy and fusion (ACDF).

Category 5 — Neurosurgery

Definition: Cranial and spinal neurosurgical procedures including craniotomy for tumor resection, aneurysm treatment, and endovascular thrombectomy.

Category 6 — Mixed / Non-specific Surgery

Definition: Cohorts defined by exclusion (e.g., “non-cardiac surgery”), broad descriptors (e.g., “any surgery”), or multi-specialty surgical groups without a dominant specialty.